



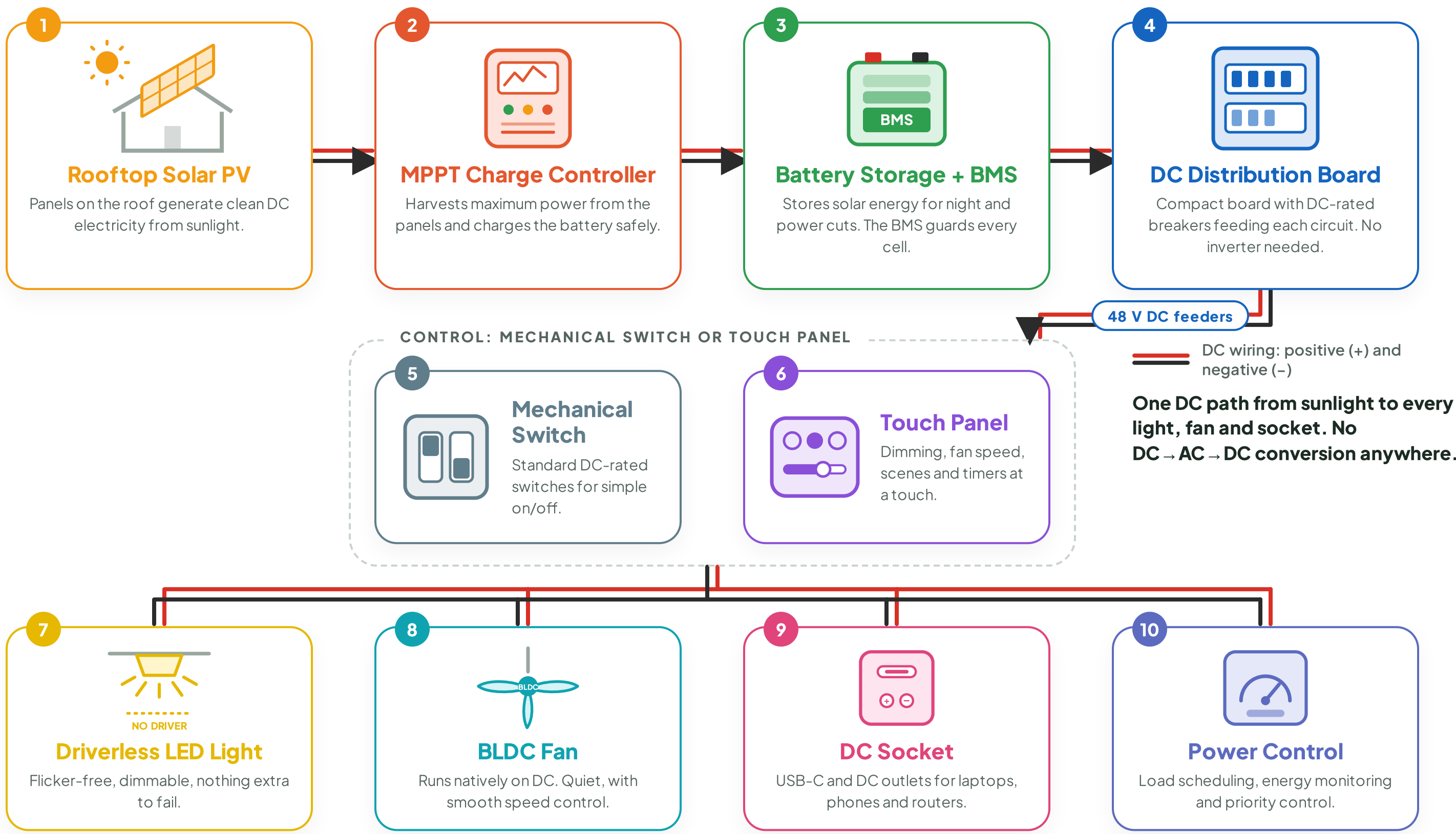
From your roof to your switch, *pure DC*

48 V
DC · EXTRA-LOW VOLTAGE

Rooftop solar charges a battery and feeds safe, low-voltage DC straight to driverless lights, BLDC fans, sockets and controls. No inverter, no conversion losses, no flicker.

How it works

Solar → storage → distribution → control → use



DC vs conventional AC

for lighting, fans and small power

ASPECT	SOLAR DC SYSTEM (48 V)	CONVENTIONAL AC (230 V)
● Safety	✓ Extra-low voltage (48 V DC). Far lower shock risk; safer for homes, schools, clinics and children.	✗ 230 V AC is hazardous to touch; relies on earthing and RCCB protection.
● Conversions	✓ Sunlight to load stays DC . No inverter, no AC-to-DC driver.	✗ DC from panels → AC via inverter → back to DC inside every LED driver, fan and charger.
● Wires	✓ Two conductors (+ and –). Extra-low-voltage circuits typically need no earth wire; short, zoned runs.	✗ Phase, neutral and earth on every circuit; long home runs from one board.
● Wire sizing	! More current at 48 V: size cables for voltage drop on long runs, or use local DC hubs. Ideal for lighting, fans and small power.	✓ Lower current, so thinner cables over long distances and for heavy loads.
● Panel	✓ Compact DC distribution board with DC-rated MCBs or fuses.	✗ Inverter, changeover switch, AC board with MCBs and RCCB.
● Circuit	✓ Simple: PV → charge controller → battery → DC board → load.	✗ More stages, more components and more points of failure.
● Light fittings	✓ Driverless LEDs : the driver, a common failure point, is gone.	✗ An AC-DC driver inside every fitting, often the first part to fail.
● Light quality	✓ Flicker-free steady light, including when dimmed.	✗ Driver ripple can cause 100 Hz flicker that tires the eyes.
● Fans	✓ BLDC fans run directly on DC, using about half the power of conventional fans.	✗ Induction fans draw around 70–80 W; BLDC models need an internal converter.
● Power cuts	✓ Battery-backed : lights, fans and sockets keep running.	✗ Grid-dependent; backup needs a separate inverter.

Benefits you can feel

10–15% Energy savings

Typical saving from removing inverter and driver conversion losses, with roughly half the power on fans using BLDC.

Longer life

Light fittings that last

No driver electronics to fail, so fittings last longer and need fewer replacements.

Flicker-free

Kind to the eyes

Steady DC light reduces eye strain, headaches and fatigue for people who work or study for long hours.

Dimming

Light for wellness

Smooth dimming for task, relaxation and circadian-friendly lighting, so spaces support how people feel.

Bigger picture

Decarbonise and *go grid-independent*

Self-generated solar power, stored and used on site, cuts reliance on a coal-heavy grid.

≈0.7 kg CO₂ avoided for every kWh of solar used instead of grid power in India

- **Lower carbon footprint** for every project, year after year
- **Resilience**: essential lights, fans and sockets stay on during outages
- **Lower peak demand** and electricity bills

Higher *IGBC ratings*

Solar DC systems can contribute towards credits across IGBC rating systems:

- **Energy efficiency**: on-site renewable energy, efficient lighting and BLDC fans
- **Indoor environment & wellness**: flicker-free light, visual comfort and lighting controls
- **Innovation**: DC distribution as an innovative design strategy

Green Interiors

Green Homes

New Buildings

Existing Buildings

Factories

Schools

Healthcare

Plan a solar DC system for your project. *Let's talk.*

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